
Milestone 2 Detailed Description

Needs Finding and Initial Concept

In milestone 2, you will perform a set of activities to establish a direction for your design.

First, you will conduct the needs finding study you designed in Milestone 1 to deepen your understanding of your users, their needs, and their current practices. Based on this improved understanding, you will begin to sketch possible designs that will address the needs you have identified. In addition, you will create personas and scenarios to represent the users who are the focus of your design and the key tasks you intend to support. Finally, you will conduct a competitive analysis to identify similar products in the marketplace and characterize their strengths and weaknesses.

Step	Activities	Output
Conduct needs finding and analysis	• Interviews • Observations • Affinity diagramming • Synthesis and prioritization	Needs finding report
Sketch and ideate	• Sketching • More sketching • Brainstorming • Selecting and refining	Design concept Storyboards
Create personas and scenarios	• Identify key user groups • Identify key tasks • Creative writing	Personas and Scenarios
Do a competitive analysis	• Search web/App stores for similar products • Identify key competitors • Map competitor functionality • Identify competitor target users • Investigate user reception (product reviews, forums)	Competitive matrix Design recommendations

Activity Descriptions

Activity 1: Conduct needs finding and analysis

Conduct needs finding and analysis	• Interviews • Observations • Affinity diagramming • Synthesis and prioritization	Needs finding report
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The core of the study will consist of interviews, potentially along with observations. You need to interview at least 10 target users to understand their needs and current practices related to the problem your project is focused on. Interviews and observations should be recorded, and the recordings should be reviewed thoroughly to extract key points.

Use affinity diagramming to synthesize the individual key points extracted from interviews and observations, and to group them into larger themes. From your affinity diagramming, you should be able to produce:

- A detailed description of users' needs
- A detailed description of users' current practices in addressing those needs
- An outline of "functional requirements", i.e. the things that your system design must accomplish in order to satisfy users' needs
- An outline of "constraints", i.e. things that your system design should avoid in order to work well and be accepted by your target users

Video on how to create Affinity Diagram: https://www.youtube.com/watch?v=R_vLlaUYHSo

Activity 2: Sketch and Ideate

Sketch and ideate	• Sketching • More sketching • Brainstorming • Selecting and refining	Design concept Storyboards
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Armed with your improved understanding of your users, their needs, and the requirements for your design, you will now unleash your creativity to come up with compelling solutions to the problems you have identified.

All of the material from the lecture "UX Design: Concept to Wireframe" is relevant to this Milestone. In that lesson, you learned that exploring multiple alternative ideas is critical to coming up with the best idea, i.e., that quantity is more important than "quality", especially in the early going. Exercise such as "10 sketches" are good starting points for making sure you don't get stuck on a particular idea too soon.

Given our short timeframe for this project, though, we won't be able to stay in brainstorming mode for long. After generating multiple ideas, sit on them for a day or two (perhaps while conducting your competitive analysis), and return to them. Referring to the functional requirements and constraints you identified in your needs finding study, along with opportunities and best practices identified in your competitive analysis, come up with tradeoff and criteria that you can use to systematically select design ideas to develop further.

After synthesizing and refining your selected design concept, create at least 3 storyboards to show

how users would interact with your design to accomplish key tasks.

Activity 3: Create Personas and Scenarios

Create personas and scenarios	• Identify key user groups • Identify key tasks • Creative writing	Personas and Scenarios
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In parallel with sketching and ideation, you will create a set of personas and scenarios to represent your users and their key tasks.

Refer to the Lesson of “UX Design: Concept to Wireframe” for pointers on how to create these artifacts. Be sure to use this activity to provide meaningful guidance to your design, choose persona and scenario details that will bring out the most important aspects of your users and their needs with respect to guiding your design. Be sure, also, to create personas that highlight differences among your user population that will be important for your design to support ((i.e., don’t create personas that need exactly the same thing to work in exactly the same way).

Create at least 2 personas, and create at least 1 scenario for each persona. These are the minimum, most substantial design projects will have more than 2 personas and scenarios.

Activity 4: Do a Competitive Analysis

Do a competitive analysis	• Search web/App stores for similar products • Identify key competitors • Map competitor functionality • Identify competitor target users • Investigate user reception (product reviews, forums)	Competitive matrix Design recommendations
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We didn’t cover competitive analysis explicitly in the class, though it is an important technique for UX Research and Design. The idea is fairly intuitive and doesn’t need a lot of explanation (thus our decision not to cover it), but some explanations and examples can be helpful, such as:

- <https://usabilitygeek.com/how-to-do-ux-competitor-analysis/>

In essence, you will use the internet to find information about other products that try to solve the same problem you are trying to solve, or problems similar to it.

You will gather as much information as you can about these products, including reading promotional materials, (e.g., what the producer of the product says about it), interacting with the product directly (if possible), and reading reviews and commentary by others (if it exists). The goal of your competitive analysis is to find out what the competition does well and where it falls short, so that you can target your own design accordingly. Ultimately your design should build upon the best practices employed by other, similar products but offer something new and different that improves upon what is already available.

A key tool in competitive matrix, This is a table that compares key competitors along important dimensions such as target users, functionality, UX style etc. Note that the article linked above focuses on market comparisons in addition to UX comparisons (e.g., “revenue streams,” “monthly

traffic”). In this project, we are not concerned with market viability, per se, so you don’t need to worry too much about these kinds of considerations. In other words, your focus should be on designing a product that would provide benefits to user and you should not worry about whether the product you design can be economically produced, maintained or marketed. While it wouldn’t be good to make crazy assumptions about such things that would defy common sense, you don’t need to be able to produce a business plan for your product as part of this course.

Milestone Deliverables

A single document (.pdf) with five clearly marked sections: Needs Finding Report, Design Concept, Personas and Scenarios, Competitive Analysis and Appendix.

Section 1: Needs Finding Report

The Needs Finding Report should include:

- A description of your target population (can be recycled from earlier reports)
- An anonymized description of your study participants, including details relevant to participants’ relationship to the target population (e.g., job title, technical expertise)
- Findings from the study, including descriptions of user needs and current practices. Use quotes and anecdotes where possible to illustrate and motivate your findings
- Implications of the study for your study design: Did you identify new requirements? Did your understanding of the users, their needs, or problems they face change? Did you identify constraints that will help steer your project away from unproductive paths?
- Also, include how you reached these findings using the affinity notes and diagramming approach. Include evidence of the process.

Section 2: Design Concept

Your Design Concept should include:

- Updated project brief, with (slightly) more detailed system description, including requirements and constraints.
- Storyboards, with captions (1 to 2 sentences explaining what’s being shown)

Section 3: Personas and Scenarios

This section should include:

- At least two personas
- At least one scenario per persona.

Section 4: Competitive Analysis

Your Competitive Analysis should include:

- A description of your selection criteria for “competitors”
- Your competitive matrix (often these are large, so you may need to reformat to make it readable within your report)
- A few paragraphs describing what you learned regarding the competitive landscape, especially best practices and opportunities for differentiation.

Appendix

You should include two items in your appendix:

- Your Needs Finding Study plan (this is to save the course instructors time tracking your plan down in case they have questions while reviewing your report)
 - If you made changes to your study plan from Milestone 1, include the revised version, along with a brief description of the changes you made
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Submission Instructions

Please submit a pdf file with the name format: HCI-Spring2026-M02-**YourTeamName**.pdf

Note: Only the placeholders in red will be replaced with your respective team name.

The pdf must be submitted on google classroom.

Make sure your Team Name, member names, roll numbers and the milestone title is clearly mentioned on the title page of the deliverable that you submit.

If your submission is late or doesn't follow the naming scheme/instructions, it will not be graded. The submission must be made on-time for it to be regarded as complete.

A note on Academic Dishonesty

All work that you submit is considered to be yours unless otherwise stated or referenced. If you consult any resources or individuals, make sure to cite them sufficiently and adequately in your submitted work. If there is any element of intentional or unintentional dishonesty e.g. Plagiarism etc found in your work, it is liable to cost you an **F** in the course.

Note: dishonesty in **any one** of the course component is sufficient to cost you an F in the entire course.